

Agentic AI — Logistics Optimization

Loaded local data: data/deliveries.csv (11 rows)

Controls

Num vehicles

5

- +

Vehicle capacity

158

- +

Num orders (destinations)

26

- +

Fuel price (per unit distance)

1.00

- +

Disruption Scenario

Random

▼

Run initial optimization

Simulate selected disruption

☒ Show map (Folium)

Dataset preview

	order_id	latitude	longitude	dema
0	DEPOT	40.8653	-73.9271	
1	ORD-1	40.7494	-73.9918	
2	ORD-2	40.7218	-73.9753	
3	ORD-3	40.7373	-73.9796	
4	ORD-4	40.7556	-73.9756	

Disruption Simulation Results

Disruption Event: Accident on Route

Agent Decision & Rationale: Severe collision detected on primary road. Re-routing to avoid standstill traffic.

Cost before

30650.00

Cost after

39089.00

↑ +8439.00

Routes (after disruption):

```
▼ [
  ▼ 0 : {
    ▼ "route" : [
      0 : 0
      1 : 5
      2 : 6
      3 : 8
      4 : 4
      5 : 3
      6 : 2
      7 : 10
      8 : 1
```

9 : 7

10 : 9

]

▼ "coords" : [

▼ 0 : [

0 : 40.86534118652344

1 : -73.92707061767578

]

▼ 1 : [

0 : 40.764591217041016

1 : -73.96166229248047

]

▼ 2 : [

0 : 40.75785827636719

1 : -73.96041107177734

]

▼ 3 : [

0 : 40.757720947265625

1 : -73.97148132324219

]

▼ 4 : [

0 : 40.75555419921875

1 : -73.97563171386719

]

▼ 5 : [

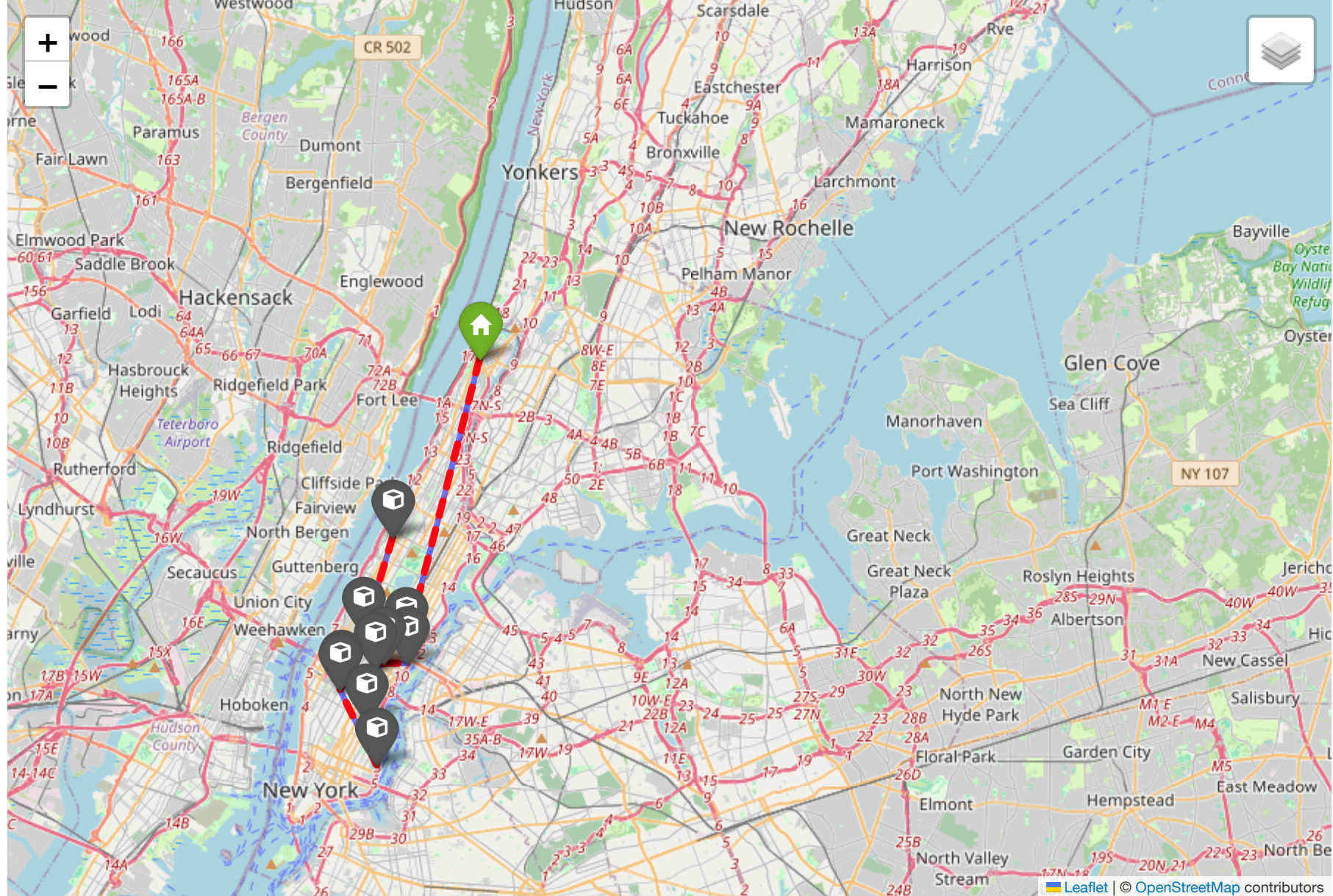
0 : 40.73733901977539

1 : -73.97962951660156

]

▼ 6 : [

```
    0 : 40.721805572509766
    1 : -73.97529602050781
  ]
  7 : [
    0 : 40.74822998046875
    1 : -73.9925308227539
  ]
  8 : [
    0 : 40.74940490722656
    1 : -73.99177551269531
  ]
  9 : [
    0 : 40.768123626708984
    1 : -73.98106384277344
  ]
  10 : [
    0 : 40.80255126953125
    1 : -73.96768951416016
  ]
]
}
```



Route Analysis (Before Disruption)

Vehicle 1 Details

Sequence: 0 -> 5 -> 6 -> 8 -> 4 -> 3 -> 2 -> 10 -> 1 -> 7 -> 9

Route Distance

16.90 miles

Route Analysis (After Disruption)

Vehicle 1 Details

Sequence: 0 -> 5 -> 6 -> 8 -> 4 -> 3 -> 2 -> 10 -> 1 -> 7 -> 9

Route Distance

16.90 miles

Cost

40000

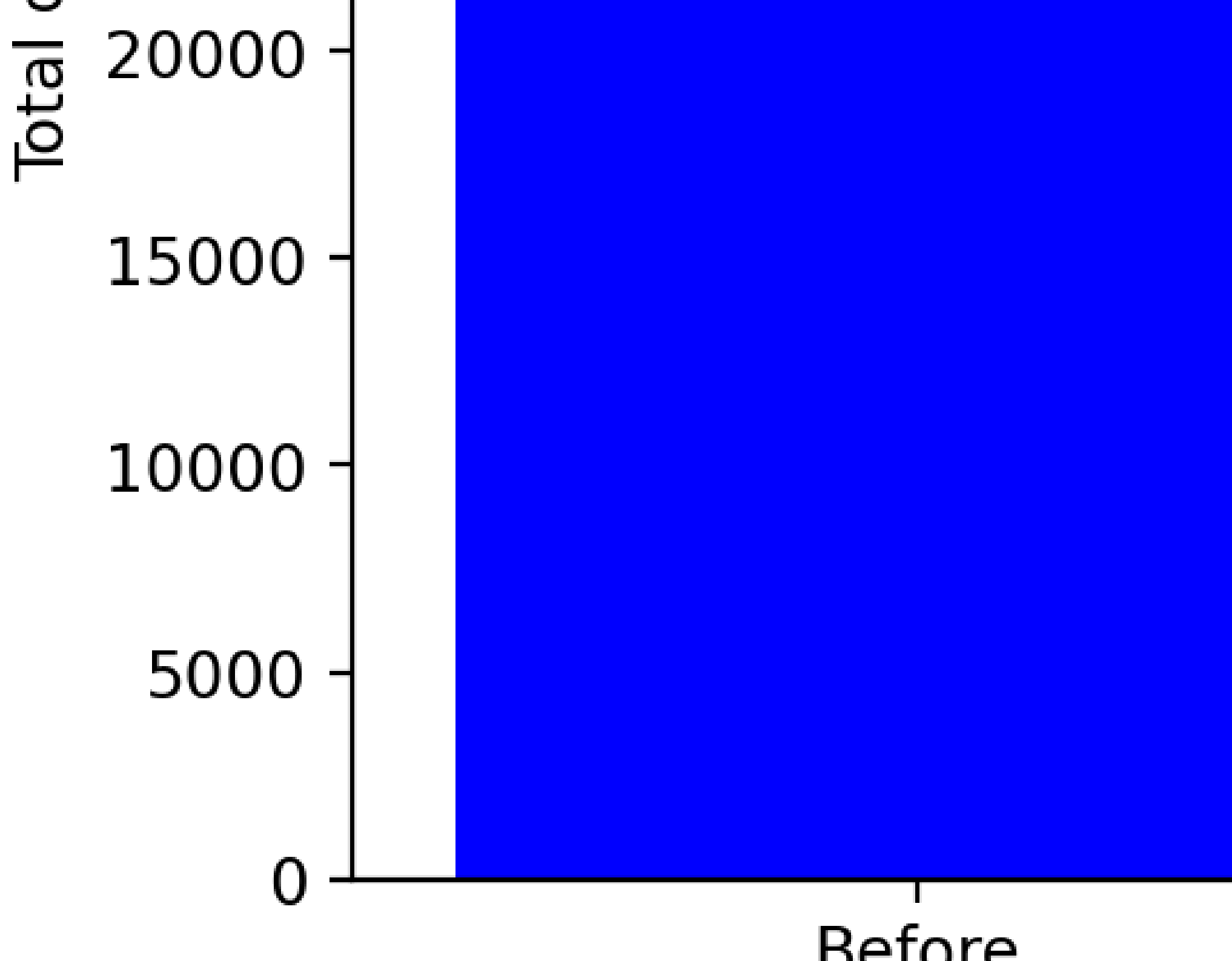
35000

30000

25000

cost





Notes: This system uses Folium/OpenStreetMap for maps to avoid paid APIs. Ensure a production Kaggle dataset is placed at data/deliveries.csv (see README)